

Immunodeficiency 70 (IMD70): A Comprehensive Disease Characteristics Report

Disease: Immunodeficiency 70 (IMD70) **Primary identifiers:** OMIM #618969 · MONDO:0033542 · DOID:0112005 · UMLS C5436501 · MedGen 1740270 **Causal gene:** *IVNS1ABP* (KLHL39 / NS1-BP), chromosome 1q25.3, HGNC:16951 **Category:** Mendelian, autosomal dominant (haploinsufficiency)

Summary

Immunodeficiency 70 (IMD70) is an **ultra-rare, autosomal-dominant inborn error of immunity** caused by **heterozygous loss-of-function (haploinsufficiency) of *IVNS1ABP***, a gene at chromosome 1q25.3 that encodes a BTB-BACK-Kelch protein (also called KLHL39 or NS1-BP). The disorder was first defined by Thaventhiran and colleagues in 2020 within the NIHR BioResource whole-genome sequencing primary-immunodeficiency cohort ([PMID: 32499645](#)), in **three unrelated probands** carrying loss-of-function alleles. It is catalogued in OMIM as #618969 and mapped to MONDO:0033542. No dedicated Orphanet, ICD-10/ICD-11, or MeSH entry exists for IMD70 as a distinct entity; clinically it falls within the common variable immunodeficiency (CVID) / combined immunodeficiency spectrum.

Clinically, IMD70 presents as a **combined immunodeficiency with immune dysregulation**. The hallmark features are **cutaneous HPV-driven warts** (verrucae, palmar and plantar warts), **recurrent bacterial sinopulmonary infections** and furuncles, **hypogammaglobulinemia**, and reduced circulating **CD4⁺ T cells and CD19⁺ B cells**. Superimposed autoinflammatory/autoimmune manifestations include colitis, celiac disease, achalasia and retinal vasculitis. Inheritance is autosomal dominant with **incomplete penetrance and variable expressivity** — a transmitting mother in the original cohort carried the mutation without an infection history yet still displayed a subclinical immunophenotype (reduced CD4⁺ T and CD19⁺ B cells).

Mechanistically, *IVNS1ABP* is a multifunctional protein that **stabilizes the F-actin cytoskeleton** through its Kelch repeats, acts as a **CUL3 E3-ubiquitin-ligase substrate adaptor**, and participates in **pre-mRNA splicing and nuclear mRNA export**. The gene is extremely intolerant to loss-of-function (gnomAD pLI $\approx$ 1.0), so a single inactivating allele reduces functional protein by roughly half and produces disease through haploinsufficiency. Diagnosis is essentially genomic (WGS/WES or a broad inborn-errors-of-immunity panel), and management is standard-of-care primary immunodeficiency support (immunoglobulin replacement, antimicrobial prophylaxis, dermatologic HPV treatment, immunomodulation, and consideration of HSCT in severe cases). No disease-specific therapy or clinical trial exists.

1. Disease Information

IMD70 is a Mendelian inborn error of immunity — a primary immunodeficiency with prominent immune dysregulation. It corresponds to **OMIM #618969**, **MONDO:0033542**, **DOID:0112005**, **UMLS C5436501** and **MedGen 1740270**. The causal gene was resolved through NCBI elink (OMIM 618969 → Gene ID 10625) as *IVNS1ABP* — Influenza Virus NS1A-Binding Protein — located at chromosome **1q25.3**. Gene synonyms/aliases include **KLHL39**, **NS1-BP**, **NS1BP**, **ARA3**, **FLARA3**, **HSPC068**, **ND1**, and — tellingly — "**IMD70**" itself. The NCBI RefSeq gene summary explicitly states the gene is "Implicated in immunodeficiency 70."

There is **no distinct Orphanet, ICD-10/ICD-11 or MeSH identifier** for IMD70; MONDO cross-references list only OMIM:618969, DOID:0112005, UMLS:C5436501 and MedGen:1740270. Clinically the entity is coded under the CVID / combined-immunodeficiency umbrella (ICD-10 D83/D81; ICD-11 4A00.x).

Source of information: the disease-level knowledge is derived from an **aggregated disease-level cohort** rather than individual EHR — specifically the whole-genome sequencing PID cohort of Thaventhiran et al., which is the source of all HPO annotations for OMIM:618969.

"Primary immunodeficiency (PID) is characterized by recurrent and often life-threatening infections, autoimmunity and cancer; and it poses major diagnostic and therapeutic challenges." — PMID: 32499645

Suggested ontology term: MONDO:0033542.

2. Etiology

Disease causal factor: IMD70 is **monogenic and genetic** — heterozygous loss-of-function of *IVNS1ABP* acting through **haploinsufficiency**. There is no infectious or environmental cause of the disorder itself (though HPV and other infections are downstream clinical consequences).

Genetic risk factors: The causal variants are germline heterozygous LoF alleles in *IVNS1ABP* (see Section 4). Because the gene is extremely LoF-constrained (gnomAD pLI $\approx$ 0.9997), essentially any inactivating allele is a strong candidate driver. No separate susceptibility loci or modifier genes have been established for IMD70.

Environmental risk factors / protective factors / gene–environment interactions: **Not established.** No environmental risk factors, protective alleles, protective exposures, or documented gene–environment interactions specific to IMD70 have been reported — consistent with a highly penetrant (though incompletely penetrant) monogenic disorder defined in only three families. This is a genuine knowledge gap rather than a negative finding.

3. Phenotypes

The HPO annotation set for OMIM:618969 comprises **18 terms**, all derived from the founding cohort (~4 patients). They cluster into four coherent domains. Onset in the described probands was **adult** (ages 19–56 at report), though a childhood contribution cannot be excluded given warts.

Phenotype (type)	HPO term	Frequency in cohort	Notes
Verrucae — cutaneous sign	HP:0200043	Characteristic	HPV-driven
Palmar warts — cutaneous sign	HP:0033004	Characteristic	HPV-driven
Plantar warts — cutaneous sign	HP:0033005	Characteristic	HPV-driven
Recurrent sinusitis — clinical	HP:0011108	Characteristic	Bacterial
Furuncle — cutaneous sign	HP:0020083	Variable	Bacterial
Immunodeficiency — clinical	HP:0002721	Characteristic	Combined
Colitis — clinical	HP:0002583	Variable	Immune dysregulation
Celiac disease — clinical	HP:0002608	Variable	Autoimmune
Achalasia — clinical	HP:0002571	Variable	Autoimmune/dysmotility
Retinal vasculitis — clinical	HP:0025188	Variable	Autoimmune
↓ CD4 ⁺ T-cell proportion — lab	HP:0032218	1/4	Immunophenotype
↓ Total B-cell count — lab	HP:0010976	1/4	Immunophenotype
↓ Circulating IgA — lab	HP:0003460	1/3	Hypogammaglobulinemia
↓ Circulating IgM — lab	HP:0002850	1/3	Hypogammaglobulinemia
↓ Total IgG — lab	HP:0032132	1/3	Hypogammaglobulinemia
↓ Circulating immunoglobulin — lab	HP:0004313	—	Hypogammaglobulinemia
Chronic fatigue — constitutional	HP:0012432	Variable	—
Autosomal dominant inheritance	HP:0000006	—	Mode

Severity/progression: variable and chronic; the immunodeficiency is lifelong. **Quality-of-life impact:** not formally measured with EQ-5D/SF-36/PROMIS instruments for IMD70; qualitatively, recurrent infections, persistent warts, chronic fatigue and gastrointestinal/ocular autoimmune complications would be expected to impair daily functioning.

"about 25% of patients have autoimmune disease, allergy is prevalent and up to 10% develop lymphoid malignancies" — PMID: 32499645

This documents the immune-dysregulation and malignancy susceptibility of the broader PID cohort in which IMD70 was described, consistent with the colitis, celiac disease and retinal vasculitis annotated to IMD70.

4. Genetic / Molecular Information

Causal gene: *IVNS1ABP* (Gene ID 10625; HGNC:16951; OMIM gene 609209), chromosome 1q25.3, reference transcript NM_006469.5, protein UniProt Q9Y6Y0.

gnomAD constraint (ENSG00000116679; GRCh38 chr1:185,296,388–185,317,273):

Metric	Value	Interpretation
pLI	0.9997	Near-certain haploinsufficient
LOEUF (oe_lof upper)	0.442	Strong LoF constraint
Observed/Expected LoF	0.317 (25 obs / 78.9 exp)	~68% depletion of LoF variants
LoF Z	5.15	Highly significant constraint
Missense Z	4.23	Missense-constrained

Pathogenic variants — ClinVar lists **three Pathogenic variants** classified to IMD70, all loss-of-function, all germline:

Variant (cDNA)	Protein	Type	Consequence
c.1899G>A	p.Trp633Ter	Nonsense	Loss of function
c.1072C>T	p.Arg358Ter	Nonsense	Loss of function
NC_000001.10:g.185276239_185287961del (~11.7 kb)	—	Intragenic/ partial-gene deletion	Loss of function

No "likely pathogenic" entries exist. IMD70-linked missense alleles (p.Arg204Cys, p.Gln504Pro, p.Asp580Ala, p.Val529Gly) remain **VUS** per ACMG/AMP; p.Cys508Gly is likely benign. Allele frequencies of the pathogenic variants are effectively absent from population databases (consistent with the LoF depletion above). **Functional consequence: loss of function / haploinsufficiency** (~50% reduction in functional protein).

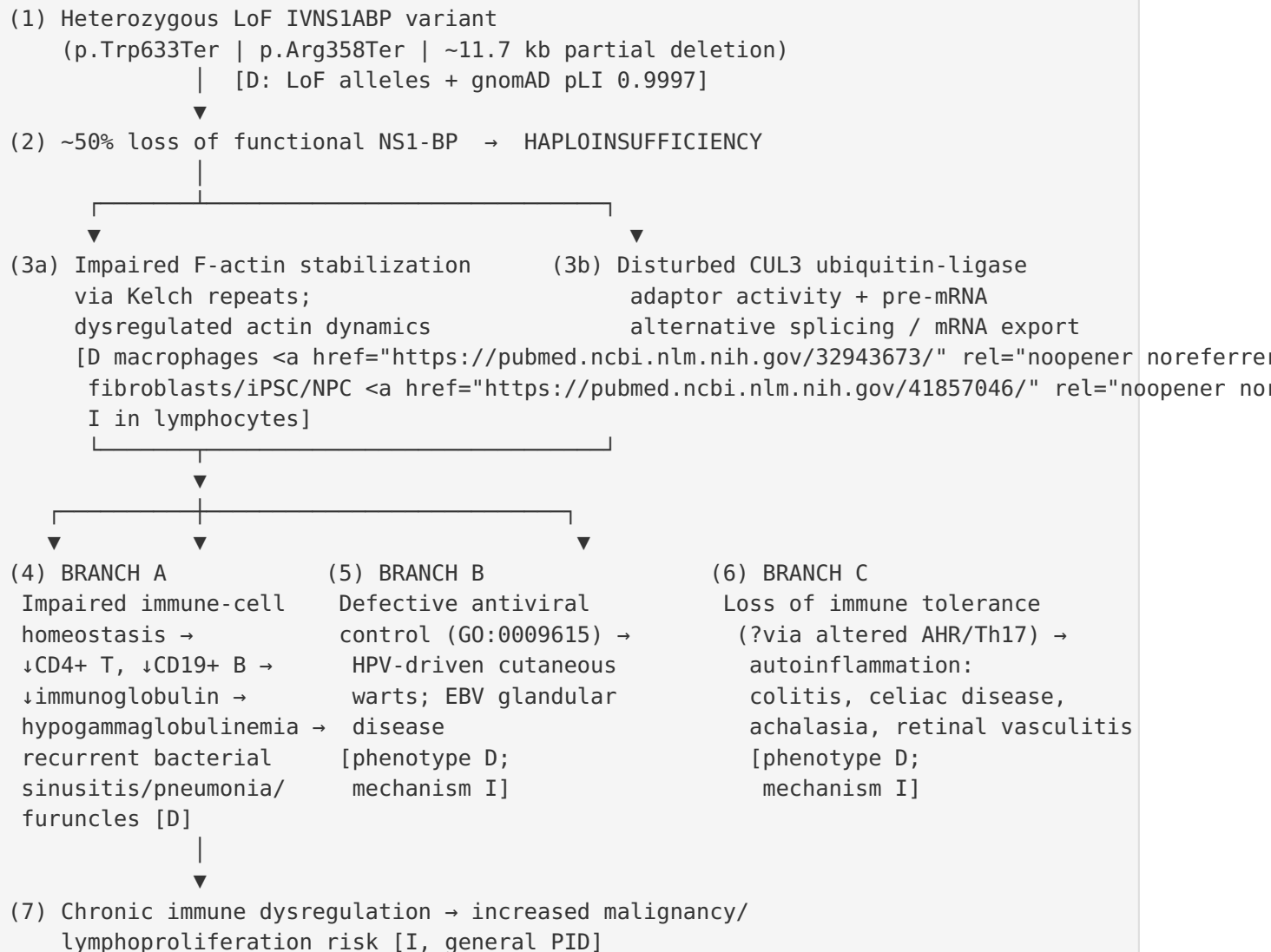
Modifier genes / epigenetic information / chromosomal abnormalities: No IMD70-specific modifier genes or epigenetic mechanisms are established. Of interest, *IVNS1ABP* protein levels are indirectly regulated epigenetically in other contexts — in liver cancer, promoter hypermethylation-driven silencing of the RNA methyltransferase NSUN7 destabilizes the *CCDC9B* transcript and reduces IVNS1ABP protein ([PMID: 37173708](#)) — but this is not shown to operate in IMD70. No recurrent chromosomal abnormality beyond the intragenic partial-gene deletion is reported.

5. Environmental Information

No environmental, lifestyle, or infectious **causes** of IMD70 have been identified — the disorder is monogenic. Infectious agents are relevant only as **downstream opportunistic consequences**: cutaneous **human papillomavirus (HPV)** driving warts, pyogenic bacteria causing recurrent sinusitis/pneumonia/furuncles, and (in the broader PID context) EBV-associated disease. There are no reported toxin, radiation, occupational, dietary, smoking, or alcohol associations specific to IMD70.

6. Mechanism / Pathophysiology

Ordered causal chain (D = demonstrated, I = inferred)



Interpretation. One inactivated *IVNS1ABP* allele reduces functional NS1-BP by ~50%. Because the gene is exquisitely dosage-sensitive ($pLI \approx 1.0$), this haploinsufficiency simultaneously perturbs three cellular systems that immune cells depend on for development, migration, antigen handling and antiviral defense:

- **Molecular pathways / cellular processes:** F-actin cytoskeleton organization (GO:0007015), the CUL3-RING ubiquitin-ligase pathway (GO:0031463), RNA splicing (GO:0008380) and nuclear mRNA export, plus modulation of the aryl-hydrocarbon-receptor (AHR) pathway

and ERK signalling. NS1-BP is a Kelch-family stabilizer of F-actin that "protects against actin derangement."

- **Protein dysfunction:** Truncating alleles/partial deletion abolish protein output from one allele; the retained protein is structurally normal but present at reduced dose (loss-of-function, not dominant-negative — although a dominant-negative contribution cannot be formally excluded).
- **Immune system involvement:** Combined immunodeficiency (reduced CD4⁺ T and CD19⁺ B cells, hypogammaglobulinemia) plus immune dysregulation/autoimmunity.
- **Antiviral defense:** Intrinsic "response to virus" role (GO:0009615) plausibly underlies the HPV-driven wart susceptibility.

"Influenza virus NS1A-binding protein (Ivns1abp) belongs to the Kelch family of proteins that play a central role in actin cytoskeleton dynamics by directly associating with F-actin and by protecting against actin derangement." — PMID: 32943673

"the cellular Non-Structural protein 1 (NS1)-binding protein (NS1-BP) interacts with the viral NS1 and M1 mRNA to promote M1 to M2 splicing" — PMID: 39384042

Molecular profiling context. Single-cell splicing-QTL analysis of PBMCs shows *IVNS1ABP* acts as a *trans*-regulator of splicing (distal effect on RPS24 splicing, interacting with the splicing factor HNRNPK), reinforcing that its dosage shapes immune-cell splicing programs (PMID: 41672992).

Upstream vs downstream. Steps 1–3 are upstream molecular lesions; steps 4–7 are downstream clinical branches. The branches are not mutually exclusive — one patient can show infection susceptibility, warts and autoinflammation together because they share the same upstream defects.

Cell types (CL): CD4⁺ T cell (CL:0000624), B cell (CL:0000236), macrophage (CL:0000235), keratinocyte (CL:0000312). **Biological processes (GO BP):** actin filament organization (GO:0007015), RNA splicing (GO:0008380), defense response to virus (GO:0051607), protein ubiquitination.

7. Anatomical Structures Affected

Organ / system level: immune system (primary); skin (warts, furuncles); upper and lower respiratory tract (sinuses, lungs); gastrointestinal tract (colon, esophagus, small bowel in celiac disease); eye (retina).

Tissue / cell level: lymphoid cells (CD4⁺ T cells, CD19⁺ B cells), macrophages, and epithelial keratinocytes are the key affected populations. Actin-dependent processes in these cells are central.

Subcellular level (GO CC): cytoskeleton (GO:0005856), Cul3-RING ligase complex (GO:0031463), spliceosomal complex (GO:0005681), nucleoplasm (GO:0005654), cytoplasm/cytosol.

Localization (UBERON): epidermis (UBERON:0001003), paranasal sinus (UBERON:0002100), lung (UBERON:0002048), colon (UBERON:0001155), esophagus (UBERON:0001043), retina (UBERON:0000966). Manifestations (e.g., warts, retinal vasculitis) are generally bilateral/multifocal rather than lateralized.

8. Temporal Development

Onset: In the defining cohort, probands presented in **adulthood** (ages 19–56 at report), consistent with the general PID observation that most patients present in adulthood without an apparent family history. Warts may begin earlier. Onset is **insidious/chronic** rather than acute.

Progression: chronic and lifelong, with a fluctuating course driven by recurrent infections and episodic autoinflammatory flares (colitis, retinal vasculitis). No defined staging system exists.

Critical periods / remission: No spontaneous remission is described; the underlying immunodeficiency is permanent. The relevant intervention window is essentially from diagnosis onward (immunoglobulin replacement, prophylaxis).

"Although the most severe forms of PID are identified in early childhood, most patients present in adulthood, typically with no apparent family history and a variable clinical phenotype of widespread immune dysregulation" — PMID: 32499645

9. Inheritance and Population

Epidemiology: Ultra-rare — defined in only **three unrelated probands** worldwide; prevalence unknown and unquantified (< **1 per 1,000,000**). No incidence, sex-ratio or geographic-distribution data exist.

Inheritance: Autosomal dominant (HP:0000006) via haploinsufficiency.

Penetrance/expressivity: Incomplete penetrance, variable expressivity. The transmitting mother in kindred A carried the mutation and lacked an infection history yet still showed reduced CD4⁺ T and CD19⁺ B cells (subclinical immunophenotype).

Other genetic-etiology parameters: No genetic anticipation, germline mosaicism, founder effect, consanguinity association, or defined carrier frequency is established (unsurprising for a dominant disorder with only three families). No population enrichment is known.

10. Diagnostics

Diagnostic approach: genomic. IMD70 was discoverable only through unbiased genome-wide sequencing (it was found within a whole-genome sequencing PID cohort) and is diagnosed by identifying a heterozygous LoF *IVNS1ABP* variant via **WGS, WES, or a broad primary-immunodeficiency / inborn-errors-of-immunity gene panel** (reference transcript NM_006469.5; NIH GTR condition C5436501). Chromosomal microarray may detect the ~11.7 kb partial-gene deletion.

Supportive laboratory work-up: serum immunoglobulins (IgG/IgA/IgM — may show hypogammaglobulinemia, HP:0004313); lymphocyte immunophenotyping (reduced CD4⁺ T cells HP:0032218, reduced CD19⁺ B cells HP:0010976); vaccine-response/antibody-function testing; HPV/EBV assessment.

Clinical criteria / differential diagnosis: No standalone diagnostic criteria exist; IMD70 is diagnosed genotype-first within the CVID/CID framework. Differential diagnoses include CVID, combined immunodeficiency, WHIM syndrome (CXCR4), other "wart" PIDs (EVER1/2/TMC6-8 epidermodysplasia verruciformis, GATA2, DOCK8, WILD/CIB1), and actin-related PIDs (WAS, ARPC1B, DOCK8).

Screening: No newborn/carrier screening program exists; cascade genetic testing of relatives is appropriate once a familial variant is identified.

"The implementation of whole-genomic analyses in the routine diagnostics has led to a paradigm shift. Upfront genome-wide analysis by whole genome sequencing (WGS) will shorten the time to diagnosis..." — PMID: 39381601

11. Outcome / Prognosis

No formal survival, mortality, or quality-of-life data specific to IMD70 exist given the tiny cohort. Prognosis is inferred from the CVID/combined-immunodeficiency framework: **chronic lifelong disease** with morbidity from recurrent infections, persistent HPV warts, autoinflammatory complications (colitis, retinal vasculitis, achalasia), and an elevated **long-term risk of lymphoproliferation/malignancy** typical of dysregulated PID. With immunoglobulin replacement and infection prophylaxis, life expectancy is likely substantially improved, though unquantified. Prognostic biomarkers specific to IMD70 have not been defined; degree of hypogammaglobulinemia and T/B-cell cytopenia are plausible severity indicators.

"about 25% of patients have autoimmune disease, allergy is prevalent and up to 10% develop lymphoid malignancies" — PMID: 32499645

12. Treatment

There is **no disease-specific therapy and no IMD70 clinical trial** (ClinicalTrials.gov). Management is **inferred standard-of-care PID care**:

Intervention	Rationale	NCIT suggestion
Immunoglobulin replacement therapy	For hypogammaglobulinemia (HP:0004313)	Immunoglobulin Therapy
Antimicrobial prophylaxis	Recurrent bacterial infections	Antibiotic Therapy
Dermatologic/ablative HPV treatment + HPV vaccination	Cutaneous warts	Human Papillomavirus Vaccine
Immunomodulation	Autoinflammatory features (colitis, retinal vasculitis)	Immunomodulatory Therapy
Malignancy/lymphoproliferation surveillance	Elevated PID malignancy risk	—
Allogeneic HSCT (consideration)	Severe combined immune dysregulation	Hematopoietic Stem Cell Transplantation

No approved **gene or cell therapy** exists. No pharmacogenomic guidance specific to IMD70 is established.

13. Prevention

There is no primary prevention for this monogenic disorder. Relevant measures are:

- **Secondary/tertiary prevention:** early genomic diagnosis, immunoglobulin replacement, infection prophylaxis, HPV vaccination, and surveillance for autoimmune complications and malignancy.
- **Genetic counseling:** autosomal-dominant recurrence risk (50% to offspring of an affected carrier), with counseling on **incomplete penetrance and variable expressivity**; cascade testing of at-risk relatives; options for prenatal/preimplantation genetic testing once a familial variant is confirmed.
- **Public-health/environmental interventions:** not applicable.

14. Other Species / Natural Disease

IVNS1ABP is deeply conserved across mammals, with one-to-one orthologs in **mouse** (*Ivns1abp*, NCBI GeneID 117198) and **rat** (*Ivns1abp*, GeneID 289089); the Kelch β -propeller is conserved broadly (orthologs and functional homologs extend to fish, where *ivns1abp* is studied as an egg-quality/ageing marker — PMID: 41270604). **No naturally occurring animal disease (OMIA) corresponding to IMD70** and no zoonotic/cross-species transmission are applicable — the disorder is a germline human Mendelian condition.

15. Model Organisms

Disease-relevant experimental systems reported to date are **cellular/in vitro, not whole-animal IMD70 models**:

1. **Patient-derived fibroblasts, isogenic iPSCs, neural progenitor cells and cerebral organoids** carrying a homozygous *IVNS1ABP* mutation (progeroid-neuropathy context) show defective cytokinesis, increased DNA damage, premature cellular senescence and dysregulated actin polymerization (PMID: 41857046).
2. **Macrophage models** showing *Ivns1abp* controls actin-dependent phagocytosis and a c-Myc-regulated reparative phenotype (PMID: 32943673).
3. A **crystal structure** of the human Kelch domain (residues 330–642) solved at 1.98 Å as a six-bladed β -propeller (PMID: 29497022).

"Exome sequencing revealed a homozygous mutation in the IVNS1ABP gene, which encodes IVNS1ABP, an influenza virus non-structural protein-1 binding protein." — PMID: 41857046

No published *Ivns1abp*-knockout mouse specifically modeling the human immunodeficiency was identified, and no natural animal model exists — a significant resource gap. Available models are best suited to studying the actin, cytokinesis, senescence and splicing arms of the mechanism rather than the intact immune phenotype.

Mechanistic Model / Interpretation (Synthesis)

The unifying model is that **a single inactivating *IVNS1ABP* allele halves NS1-BP dosage in a gene that cannot tolerate loss-of-function**, and this dosage insufficiency degrades three cellular systems at once — the F-actin cytoskeleton, the CUL3-ubiquitin proteostasis machinery, and the mRNA-splicing/export program. Immune cells are unusually dependent on all three (for immune-synapse formation, migration, phagocytosis, antigen processing and antibody production), which is why the clinical picture is a **combined immunodeficiency with immune dysregulation** rather than an isolated defect. The three downstream branches (infection susceptibility, HPV-driven warts, autoinflammation) coexist because they emanate from the same upstream lesion.

The strongest mechanistic evidence is in the **actin arm** — NS1-BP is a Kelch-family F-actin stabilizer ([PMID: 32943673](#)), and defects in other actin regulators (WASP, ARPC1B) are established causes of immunodeficiency ([PMID: 29127144](#)). The **splicing/export arm** is well established biochemically ([PMID: 39384042](#)), and immune-cell splicing-QTL data confirm *IVNS1ABP* dosage regulates splicing programs *in trans* ([PMID: 41672992](#)). The connection from these molecular defects to the specific human lymphocyte phenotype remains **inferred rather than directly demonstrated in patient T/B cells**.

Evidence Base

PMID	Title (abbrev.)	Role in this report
32499645	<i>Whole-genome sequencing of a sporadic primary immunodeficiency cohort</i>	Foundational. Defines IMD70; source of all HPO annotations; establishes AD inheritance, adult presentation, variable dysregulation, malignancy risk
32943673	<i>The influenza virus NS1A binding protein gene modulates macrophage response...</i>	NS1-BP as Kelch-family F-actin stabilizer; actin/phagocytosis mechanism
39384042	<i>Cellular NS1-BP interacts with mRNA export receptor NXF1...</i>	Splicing (M1→M2) and NXF1-mediated mRNA-export functions
41857046	<i>IVNS1ABP mutation drives cellular senescence in newly identified progeroid neuropathy</i>	Biallelic (recessive) phenotype; zygosity-dependent disease; senescence/DNA-damage cellular models
29497022	<i>Crystal structure of the Kelch domain of human NS1-binding protein at 1.98 Å</i>	Protein architecture — six-bladed β -propeller Kelch domain
41672992	<i>Single-cell resolution of splicing regulation in PBMCs...</i>	<i>IVNS1ABP</i> is a <i>trans</i> -sQTL regulator of splicing in immune cells (dosage sensitivity)
39381601	<i>Rapid identification of primary atopic disorders by upfront genomic sequencing</i>	Supports genome-first diagnostic paradigm for ultra-rare monogenic immune disease
29127144	<i>Disruption of thrombocyte and T-lymphocyte development by <i>ARPC1B</i> mutation</i>	Comparator: actin-regulator (Arp2/3) deficiency causing PID; supports actin-immunodeficiency axis
37173708	<i>Epigenetic inactivation of <i>NSUN7</i>... in liver cancer</i>	Shows <i>IVNS1ABP</i> protein can be indirectly downregulated via epigenetic/RNA-modification pathways (non-IMD70 context)

Additional papers describe *IVNS1ABP* in cardiovascular/oncologic contexts (PMID: 40782973, PMID: 41613752, PMID: 38956669) and non-mammalian biology (PMID: 41270604, PMID: 39664389); they corroborate the gene's broad roles in actin dynamics, MYC regulation and cell-death/senescence programs but do not bear directly on IMD70 pathogenesis.

Limitations and Knowledge Gaps

1. **Extremely small evidence base.** The disease definition rests on **three unrelated probands** from a single 2020 study; all HPO annotations trace to one publication. Prevalence, incidence, sex ratio, penetrance estimates and genotype-phenotype correlations are effectively unquantifiable.
2. **Mechanism inferred in lymphocytes.** NS1-BP's actin and splicing functions are demonstrated in macrophages, fibroblasts, iPSCs and biochemical systems — not in patient T or B cells. Branches A–C are inferred.
3. **No animal model of the immunodeficiency.** No *Ivns1abp* heterozygous/conditional-knockout mouse modeling IMD70 exists; no OMIA natural-disease counterpart.
4. **VUS burden.** Beyond three LoF alleles, IMD70-associated missense variants remain VUS, limiting diagnostic certainty for non-truncating variants.
5. **No IMD70-specific therapeutics or trials.** Management is extrapolated from general PID/CVID practice; no efficacy data specific to this disorder.
6. **Ontology/coding gaps.** No Orphanet, ICD-10/11, or MeSH identifier — a barrier to registry-based epidemiology.
7. **Zygosity-dependent phenotypes.** Biallelic *IVNS1ABP* loss causes a distinct progeroid neuropathy; the relationship (if any) to the heterozygous immunodeficiency spectrum is unexplored.

Proposed Follow-up Experiments / Actions

1. **International case ascertainment** via GeneMatcher/matchmaker exchanges and inborn-errors-of-immunity consortia to refine penetrance, expressivity and the phenotypic spectrum.

2. **Patient-derived immune-cell functional studies** — profile CD4⁺ T, CD19⁺ B and keratinocyte lines for F-actin content, immune-synapse formation, migration, phagocytosis and antibody class-switching to test branches A and B directly.
3. **Isogenic haploinsufficiency models** — heterozygous *IVNS1ABP* iPSC lines differentiated to lymphoid/myeloid lineages and skin organoids to quantify dosage-dependent effects on actin, splicing and ubiquitination.
4. **Conditional mouse model** — lymphocyte-specific *Ivns1abp* heterozygous/knockout mouse to test whether haploinsufficiency recapitulates cytopenias, hypogammaglobulinemia and infection susceptibility.
5. **Single-cell RNA-seq of patient PBMCs** to map mis-splicing events (e.g., RPS24) driven by reduced *IVNS1ABP* dosage, building on [PMID: 41672992](#).
6. **HPV/keratinocyte antiviral assays** to test whether NS1-BP–haploinsufficient keratinocytes have impaired intrinsic antiviral control explaining the wart phenotype.
7. **Registry and ontology curation** — advocate for an Orphanet entry and ICD-11 extension code to enable systematic epidemiology.
8. **Standardized management guidance** — draft a consensus care pathway (Ig-replacement thresholds, HPV surveillance, malignancy screening, HSCT criteria) tailored to IMD70 within the CVID/CID framework.

Report compiled from a 5-iteration autonomous investigation: 9 confirmed findings, 14 papers reviewed. Evidence source types span human clinical cohort

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